SITE AUTOMATION

Reports on a file share

Every report job's page written straight onto
a Windows file share over SMB - no mount

Site Automation - set up, test and use | site-automation 0.99.0 | October 2026

What this is. Every report job can write its report as a web page into a folder on a Windows file share, for example `\\fileserver\its\aap\reports`, straight over SMB: nothing mounted on the AAP execution nodes, no AAP setting, no web server that takes uploads. This guide sets it up from nothing, in order: the folder, the account, the execution environment, the network, the credential, the settings, a test - and your reports page.

How to read it
Steps 1 to 8 in order. The table at the end says what each message means. Rendered from `docs/REPORT_SHARE.md` in the site-automation repository (version 0.99.0); copy YAML from the repository, not from the PDF.

Host names and addresses in this guide are placeholders. Copy YAML from the repository files, not from the PDF.

Contents

1. The reports on a file share (SMB) 3

What you get . . . 3
What you need . . . 3
Step 1. The folder and its permissions . . . 4
Step 2. The service account . . . 4
Step 3. The execution environment . . . 4
Step 4. The network path . . . 5
Step 5. The credential in AAP . . . 5
Step 6. The settings . . . 5
Step 7. Test it . . . 6
Step 8. The reports page (your portal) . . . 6
When something fails . . . 7
The other ways . . . 8

1. The reports on a file share (SMB)

Every report job can write its report as a web page into a folder on a Windows file share - for example `\\fileserver\its\aap\reports` - straight over SMB, the way Windows itself writes to a share. Nothing is mounted on the AAP execution nodes, no AAP setting changes, and no web server has to take uploads: the job logs in to the share with a service account and writes the files. A reports page (your site's portal, IIS, anything that reads that folder) then shows them.

This guide sets it up from nothing, in order. The report website itself - the names of the files, the index, the looks, the copy job - is in WEB_REPORTS.md.

What you get

After each run of a report job, in the share's folder:

```
\\fileserver\its\aap\reports\
    index.html                          every report's last run, worst first (and index.json)
    linux-health-check\
        latest.html                     always this template's newest run (and latest.json)
        2026-10-26_0605_job-1234.html   each run's own page (and its .json)
    vm-health-report\
        latest.html
        ...
```

One folder per job template (its name in AAP, lower case with dashes). `latest.html` is the file to point a reports page's tile at: it is always the newest run. Each page stands alone - its logo is inside it, it needs no script and reaches nothing on the network.

The job prints where it wrote the page (`Published: \\fileserver\its\aap\reports\linux-health-check\...`), or its web address when you set one (`Report: https://...`). A run that cannot write stops with the reason in plain words.

What you need

Item	Example	Who gives it to you
A folder on the share	`\\fileserver\its\aap\reports`	you, or the storage team
A service account that may write there	`YOURDOMAIN\svc_aap_reports`	the Active Directory team
SMB (TCP 445) from the AAP execution nodes to the file server		the network or firewall team
`smbprotocol` in the execution environment	in all three EEs of `ee/` since 0.32.0	you (step 3)
The **Report website** credential type in AAP	`aap/credential_types/report_web.yml`	you (step 5)

Step 1. The folder and its permissions

1. Make the folder once: `\\fileserver\its\aap\reports`. The job never makes it - if it is not there (a typo, a share that moved), the job stops instead of writing somewhere else. It makes the folders inside it.
2. Give the service account (step 2) **Modify** on that folder (NTFS: *Properties* > *Security*) - Modify, not Write: the job also deletes older runs' pages (`report_web_history`, docs/WEB_REPORTS.md). On the share itself it needs at least **Change** (*Properties* > *Sharing* > *Advanced Sharing* > *Permissions*) - most shares give *Everyone: Change* or *Authenticated Users: Change* already and leave the rest to NTFS.
3. Give whoever reads the reports **Read** on the folder: the reports page's account (an IIS application pool identity, a service account), and the people who open the files.

A mixed-mode (NFS and SMB) volume, as on a NetApp: ask the storage team which security style the volume (or qtree) has. **NTFS** or **mixed** is what you want here: the files the job writes over SMB then take the folder's Windows permissions, and a Windows reader can always read them. With **UNIX** security style, the files get UNIX owners and modes from the SVM's name mapping - ask for NTFS, or test that the reports page can read a file the job wrote (step 7).

Step 2. The service account

Ask the Active Directory team for an account only this uses:

```
Request: a service account for Ansible Automation Platform (AAP) to write report files to a share

- User name: svc_aap_reports (or your naming standard), in YOURDOMAIN
- Rights: Modify on \\fileserver\its\aap\reports (only there); no interactive logon (deny "Log on
  locally"
  and "Log on through Remote Desktop Services")
- Password: set, "User must change password at next logon" off. Tell me the expiry or rotation
  rule, so the
  AAP credential can be updated in time.
- Logon: SMB from the AAP execution nodes: [their names or addresses]
```

Step 3. The execution environment

The job writes over SMB with **smbprotocol**, a pure-Python SMB 2/3 client (no compiler, no Samba). It is in all three execution environments of `ee/` (Windows, network, all-in-one) from release 0.32.0: build the ones your report templates use from those files or later, and update the image in AAP - EXECUTION_ENVIRONMENT.md, chapters 3 to 5 (one per EE) and "Put an EE into AAP". Check the new image:

```shell
podman run --rm site-ee-all:3 ee-check | grep -i smb
#   YES  smbprotocol
```

Only one execution environment needs it if you let the copy job write the pages (`report_web_copy: true`, WEB_REPORTS.md, "The copy job"): the report templates keep their own execution environments and credentials, each job keeps its report in AAP, and only the template **Report website - copy** - in the EE that has smbprotocol, with the credential of step 5 - writes them to the share.

A job in an execution environment without smbprotocol stops with: *this execution environment has no smbprotocol*.

Step 4. The network path

TCP **445** from every AAP execution node that runs report jobs to the file server. Test it from a node:

shell

```
timeout 3 bash -c '</dev/tcp/fileserver.yoursite.mil/445' && echo open || echo blocked
```

`blocked`: ask the network team for TCP 445 from those nodes to the file server.

Step 5. The credential in AAP

1. **The credential type** (once): *Administration* > *Credential Types* > *Add*, name **Report website**, with the *Input configuration* and *Injector configuration* of `aap/credential_types/report_web.yml`. (It is the same type the WinRM and upload modes use: if you have it, skip this.)
2. **The credential**: *Automation Execution* > *Infrastructure* > *Credentials* > *Add*, type **Report website**, name e.g. `Report share`, **User name** `svc_aap_reports@YOURDOMAIN.MIL` - the account, `@`, and the domain's full name in capitals (its Kerberos realm) - and **Password**. Leave *Upload token* empty. `YOURDOMAIN\svc_aap_reports` works only with NTLM, never in a FIPS execution environment (step 6).
3. **Attach it** to each report template (*Templates* > *the template* > *Edit* > *Credentials*), next to its other credentials - or only to **Report website - copy**, with the copy job.

Step 6. The settings

For every report job at once, in `playbooks/group_vars/all.yml` (the report jobs read them there, whatever their inventory):

single quotes: the backslashes stay; the server by its full name - Kerberos asks for a ticket to that name

```
report_web_dest: '\\fileserver.yourdomain.mil\its\aap\reports'
# negotiate (Kerberos, else NTLM) - leave it, FIPS too; ntlm; kerberos (only a ticket already in
  the job)
report_web_smb_auth: negotiate
# encrypt every byte (SMB 3); a share that requires it is encrypted anyway
# report_web_smb_encrypt: true
# only if the folder is also served at a web address: the emails then link it
# report_web_url: https://reports.yoursite.mil/aap
```

`smb://fileserver.yourdomain.mil/its/aap/reports` works too. Not in an inventory group's Variables: the page is written by the AAP side of the job, which does not get a group's variables. To try it on one template first, put the lines in that template's **Variables** box.

FIPS. Leave `report_web_smb_auth: negotiate`. It logs in with Kerberos first, with the credential's password - no ticket to set up - when the user name is `svc_aap_reports@YOURDOMAIN.MIL` (step 5) and `report_web_dest` names the server by its full name. NTLM, its fallback, needs the MD4 hash, which a FIPS-mode execution environment does not have: the message *NTLM needs MD4* means Kerberos did not happen - check those two. Not `kerberos`: it sends a bare Kerberos token, and a server that answers in SPNEGO - what `negotiate` speaks - stops it with *Invalid token was supplied*. The EEs of `ee/` have the Kerberos settings it needs; to try Kerberos from one by hand: EXECUTION_ENVIRONMENT.md, EE 1, "Step 4. Check Kerberos to Active Directory".

Step 7. Test it

1. Launch one report template - **VM - health report**, say - with **Job type: Check**. Check writes nothing, but it logs in and looks at the folder. **You should see:** `Dry run (Check): logged in to fileserver, the folder \\fileserver\its\aap\reports is there; the report would be published at ...`
2. Launch it for real. **You should see:** `Published: \\fileserver\its\aap\reports\vm-health-report\2026-..._job-N.html`.
3. Open the folder in Windows Explorer: the template's folder with `latest.html`, and `index.html`. Open them in a browser.
4. As the reports page's account (or a colleague), open `latest.html` too: proof that the readers can read what the job wrote (step 1, mixed-mode volumes).

Step 8. The reports page (your portal)

1. Its report folder: `\\fileserver\its\aap\reports` (for a portal with a `reportFolders` setting in its configuration, that path).
2. One tile per report, pointing at that template's `latest.html`, e.g. `\\fileserver\its\aap\reports\vm-health-report\latest.html` - always the newest run. `index.html` makes a good "every report" tile: every template's last run, worst first, each linked.
3. A live miniature of the page is fine (the pages are plain HTML). The pages need **no scripts**: leave "run the report's own scripts" off.
4. A tile appears once the template has run once with the settings of step 6.

When something fails

The job says	Why	Do this
`this execution environment has no smbprotocol`	the template's EE is older than 0.33.0's all-in-one	build and use the new EE (step 3), or the copy job
`no login for the share ...: attach the "Report website" credential`	the template has no Report website credential	step 5.3
`the user name or password is wrong ... or the account is locked`	the login was refused	the credential's user name (`svc_aap_reports@YOURDOMAIN.MIL`) and password; is the account locked or expired?
`Configuration file does not specify default realm`	the user name has no realm (`YOURDOMAIN\user`, or no domain)	`svc_aap_reports@YOURDOMAIN.MIL`, the realm in capitals (step 5.2)
`Server not found in Kerberos database`	the server's short name in `report_web_dest`	its full name: `\\fileserver.yourdomain.mil\its\...` (step 6)
`Invalid token was supplied`	`report_web_smb_auth: kerberos`	`negotiate` (step 6, FIPS)
`the account may log in but not use this folder`	access denied	step 1: Modify on the folder, Change on the share
`no such share on that server`	the part after the server name is wrong	the share's name: `\\fileserver\its`, as Explorer shows it
`the folder ... is not there - make it once by hand`	the folder under the share is missing	make it (step 1), or correct `report_web_dest`
`cannot be reached on port 445`	firewall, routing or the name	step 4
`Message stream modified` (`SpnegoError ... Unspecified GSS failure`), now and then, on a share the job writes to all day	Kerberos's KRB_AP_ERR_MODIFIED: the ticket for the file server is sealed with a key it no longer has	Usually nothing: it passes - a NetApp SVM's (or any server's) machine password just changed and AD has not replicated it to every domain controller yet: the job tries again from a new login with a fresh ticket (`report_web_retries`, `report_web_retry_delay`) and the job goes on with a warning if it still fails (`report_web_required: false`). Every time: the name in `report_web_dest` is an alias (a DNS CNAME) or an address - use the server's own full name, or register the alias's SPN on its computer account (`setspn -S cifs/alias.yourdomain.mil SERVER`); or the same SPN is on two accounts (`setspn -X` lists duplicates)

The job says	Why	Do this
`[WARNING]: The report website: try 1 of 4 failed (...)`	a failure that passes, tried again from a new login (the row above, a connection reset, a KDC not reached, a session expired)	nothing, when a later try publishes it (`Published (after N tries)`)
`WARNING - The report website copy failed: ...; the email is the full report`	still not published after the retries, or a real error (not retried)	the job stays green and the email says it at the top; the row for its reason. `report_web_required: true` fails the job instead
`NTLM needs MD4 ... Kerberos did not happen`	a FIPS execution environment, and Kerberos did not log in	step 6, FIPS: `negotiate`, `user@YOURDOMAIN.MIL`, the server's full name
The page is there, but the reports page cannot open it	the readers lack Read, or a UNIX-style volume	step 1

The other ways

The same pages, written another way - pick one (WEB_REPORTS.md, "Where the files go"):

- **A share mounted on the execution nodes** (`report_web_dest: /mnt/reports/aap`): an NFS or SMB mount on every node, exposed to the jobs (*Settings* > *Job settings* > *Paths to expose to isolated jobs*). No credential in the job, no smbprotocol; root on the nodes and the same mount on each.
- **A Windows server over WinRM** (`report_web_host`): the pages written to a folder **on** that server (`D:\reports`) - not a share, which would be a second hop.
- **A web server that takes uploads** (HTTP PUT, WebDAV).